

FIN 740 FORMULA SHEET

- **General formula for taking present value of cash flow stream**

$$PV = C_0 + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_N}{(1+r)^N}$$

- **Present value of an annuity**

$$PV\left(\text{Annuity of } C \text{ for } N \text{ Periods with Interest Rate } r\right) = C \times \frac{1}{r} \left(1 - \frac{1}{(1+r)^N}\right)$$

- **PV of a growing annuity**

$$PV = C \times \frac{1}{r-g} \left(1 - \left(\frac{1+g}{1+r}\right)^N\right)$$

- **Present value of perpetuity**

$$PV(C \text{ in Perpetuity}) = \frac{C}{r}$$

- **PV of growing perpetuity**

$$PV\left(\text{Growing Perpetuity}\right) = \frac{C}{r-g}$$

- **Free cash flow**

$$\begin{aligned} &\text{Unlevered net income} + \text{depreciation} \\ &- \text{capital expenditures (CapEx)} \\ &- \text{change in net working capital } (\Delta\text{NWC}) \end{aligned}$$

- **Unlevered net income**

$$\text{EBIT} \times (1 - \text{Tax rate}), \text{ where EBIT} = \text{Revenues} - \text{expenses} = \text{depreciation}$$

- **After-tax salvage value from an asset sale**

$$\text{After-Tax Cash Flow from Asset Sale} = \text{Sale Price} - (\text{Tax Rate} \times \text{Capital Gain})$$

- **Net working capital**

$$\begin{aligned} \text{Net Working Capital} &= \text{Current Assets} - \text{Current Liabilities} \\ &= \text{Cash} + \text{Inventory} + \text{Receivables} - \text{Payables} \end{aligned}$$

- **Discounted cash value of an entire company (i.e., a company's asset value)**

$$Firm\ value = Cash_0 + \frac{FCF_1}{(1 + r_{wacc})} + \frac{FCF_2}{(1 + r_{wacc})^2} + \dots + \frac{FCF_{\infty}}{(1 + r_{wacc})^{\infty}}$$

- **Finding equity value & stock price**

$$Equity\ value = Firm\ value - debt$$

$$Stock\ price = Equity\ value / Shares\ outstanding$$

- **Enterprise value**

$$Enterprise\ value = Firm\ value - cash_0$$

- **Expected return and risk**

$$Expected\ return = risk\text{-}free\ rate + risk\ premium$$

- **Capital Asset Pricing Model (CAPM)**

$$Required\ return = r_f + [E(r_m) - r_f] \times \beta$$

- **Weighted Average Cost of Capital**

$$r_{WACC} = \left(\frac{E}{D + E}\right)r_e + \left(\frac{D}{D + E}\right)r_d(1 - T)$$

- **Bond valuation formula**

$$Bond\ Value = \frac{C}{r} \left[1 - \frac{1}{(1 + r)^t} \right] + \frac{F}{(1 + r)^t}$$

- **Dividend discount formula (for valuing a stock)**

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \frac{Div_3}{(1 + r_E)^3} + \dots$$

- **Terminal value formula (assuming a growing perpetuity)**

$$V_N = \frac{FCF_{N+1}}{r_{wacc} - g_{FCF}} = \left(\frac{1 + g_{FCF}}{r_{wacc} - g_{FCF}} \right) \times FCF_N$$